Bulk Canvas → Moodle Migration

A step-by-step guide for administrators

Migrating many courses off Canvas at once with the **CourseUplift** (`tool_courseuplift`) and **Automate** (`tool_automate`) Moodle admin tools.

Applies to	CourseUplift 1.3.0+ and Automate 1.4.2+, on Moodle 5.0–5.3 LTS
Audience	Site administrators and managers running a Canvas migration
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1. Overview

CourseUplift imports a single Canvas Common Cartridge (`.imscc`) export into Moodle: it can **analyse** a package (report what it contains and how cleanly each part maps to Moodle) or **build** a brand-new Moodle course from it. For a handful of courses you can drive it by hand from its own upload page.

When you are moving **many** courses, the Automate plugin drives CourseUplift in bulk. You feed it a list of Canvas backup URLs and/or a server directory of packages; Automate turns each package into one background CourseUplift job. This guide covers that bulk workflow end to end.

The two ways each package can be handled

Build courses now	A new, hidden Moodle course is created from each package automatically. Best when you trust the packages and want courses ready to review.
Analyse for later	Each package is fetched and a conversion report produced, but no course is built. An administrator reviews each report and builds the course manually from CourseUplift when ready.

Why background jobs?

Nothing is converted inside your browser request. Every package becomes a background adhoc task, so a directory or URL list of hundreds of courses is queued instantly and then worked through by cron instead of timing out the page.

CourseUplift caps how many run at once with its **Maximum simultaneous conversions** setting (default 2 analyse jobs and 2 build jobs), so a large batch never takes every cron worker and the rest of the site's background work, such as notifications, keeps flowing. Jobs beyond the limit wait in the queue and start as others finish.

2. Before you start

- **Both plugins installed.** CourseUplift provides the import engine; Automate provides the bulk driver. Automate detects CourseUplift automatically — the bulk-import feature only appears when it is present.
- **Capabilities.** You need `tool/automate:manage` to run Automate, and `tool/courseuplift:use` for the import itself and the *Conversion jobs* page. Building courses additionally needs `moodle/course:create` in the target category.
- **Cron running.** The imports run as background tasks, so Moodle cron must be active (*Site administration* > *Server* > *Tasks* > *Task processing*).
- **Trusted packages only.** Imported HTML is stored and displayed as authored. Only import `.imscc` packages from sources you trust.

Install or upgrade without locking the site

While any plugin installs or upgrades through *Notifications*, Moodle shows "Site is being upgraded, please retry later" to everyone and pauses cron until it finishes. To avoid that, run the upgrade from the command line. When the first command exits with code 3 (maintenance not required, as for these plugins), the second upgrades while users carry on working:

```
php admin/cli/upgrade.php --is-maintenance-required
php admin/cli/upgrade.php --non-interactive --maintenance=false
```

3. Turn on bulk Canvas import

The feature ships behind an off-by-default kill-switch. Enable it once:

Step 1 — Open Automate's settings

Go to *Site administration* > *Plugins* > *Admin tools* > *Automate* > *Settings*.

Step 2 — Allow bulk Canvas import

Tick **Allow bulk Canvas import**. The background import task re-checks this switch before it runs, so clearing it later also stops any still-pending bulk import from starting — but it does not cancel a CourseUplift job that has already been created, so a build already under way still finishes.

Step 3 — (Optional) set a source directory

If you will import from a folder of packages on the server, set **Bulk Canvas import source directory** to that folder (a staging area holding `.imscc` / `.zip` files). You can skip this if you only import from URLs.

Step 4 — (Optional) choose how many conversions run at once

In *Site administration* > *Plugins* > *Admin tools* > *CourseUplift*, set **Maximum simultaneous conversions**. The default of 2 suits most sites; raise it on a server with spare cron capacity to finish a large batch sooner, or lower it to 1 to leave more room for other background work. The limit applies to analyse jobs and build jobs separately.

4. Prepare your packages

Option A — A list of Canvas backup URLs

In Canvas, export each course as a Common Cartridge and collect the download URLs — one per line. Each is fetched in the background.

```
https://school.instructure.com/exports/course-101.imscc
https://school.instructure.com/exports/course-102.imscc
https://school.instructure.com/exports/course-103.imscc
```

> **Security: server-side URL fetch (SSRF)**
> When you import from a download URL, the site fetches it server-side through Moodle's SSRF-aware cURL layer, governed by *Site administration* > *Security* > *HTTP security*. On Moodle 5.0+ these settings block loopback and private ranges by default. Keep those defaults; uploading the file directly avoids the fetch entirely.

Option B — A directory of packages

Copy the exported `.imscc` files into the source directory you configured in step 3. The picker searches server-side and type-to-filters, so it scales to hundreds of files.

You can use either source, or both together in one run.

5. Run a bulk import

Step 1 — Open the bulk import page

Go to *Plugins* > *Admin tools* > *Automate* > *Bulk Canvas import*.

Step 2 — Choose your sources

Paste your URL list into the text area, and/or pick packages from the server directory. Preview the selection before queueing.

Step 3 — Choose what happens to each package

- **Build courses now** — a new course is created from each package (needs course-create permission in the target category).
- **Analyse for later** — each package is analysed only, ready for you to review and build manually.

Step 4 — Set the target category and options

Pick the course category new courses land in. Optional build settings (such as building a runnable quiz from each standalone question bank, or combining consecutive pages into a book/lesson) carry through to each job.

Step 5 — Queue the jobs and follow them

Confirm to queue every package as a background job. Follow the whole batch on CourseUplift's **Conversion jobs** page (*Site administration* > *Courses* > *Conversion jobs*) — see section 8. The tasks also appear in *Site administration* > *Server* > *Tasks* > *Task logs* as **Analyse Canvas package** and **Build course from Canvas package** (queued under **Bulk Canvas import**).

6. Review & build staged imports

Packages you queued with **Analyse for later** appear under *Automate* > *Staged Canvas imports* once analysed. From there you can:

- **Review a report and build the course** — open a completed analyse job to reach CourseUplift's status page, read the conversion report, and use **Build this course** when you are happy.
- **Jump to a built course** — a completed build links straight to the course it created.
- **See storage used** — the top of the page shows how much space your stored packages occupy.
- **Reclaim space** — tick finished imports and **Delete selected** to free their stored packages.

> **Deleting is safe**
> Deleting a finished import frees its stored `.imscc` package **once no other job still references it** (an analyse job and the build made from it share one package, so it is freed only when the last of them is deleted) but **keeps any course a build already created** — it reclaims cache, it does not undo an import. Only finished (done or failed) imports can be deleted, so a delete never interrupts an in-flight conversion.

7. Driving it from the command line

The same bulk job can run from the CLI — handy for very large migrations or scripting. Run with `--help` for all options, or `--list` to inspect the sources without queueing.

```
# Build a course from every URL in a file, into category 5:
php admin/tool/automate/cli/import_canvas.php \
    --urls=/data/urls.txt --category=5 --mode=build --execute

# Analyse a directory of packages for later review:
php admin/tool/automate/cli/import_canvas.php \
    --source=/data/canvas --category=5 --mode=analyse --execute

# Inspect what would be queued, without queueing anything:
php admin/tool/automate/cli/import_canvas.php --urls=/data/urls.txt --list

# Follow the jobs (add --json for scripts, --status=failed for failures only):
php admin/tool/courseuplift/cli/jobs.php
```

8. Monitor & troubleshoot

CourseUplift's **Conversion jobs** page (*Site administration* > *Courses* > *Conversion jobs*) lists every analyse and build job on the site:

- **Totals** — queued, running, done and failed, plus items created, items skipped and warnings.
- **Live progress** — each job's progress bar and current step, the course a build created and its counts. The page refreshes itself every 10 seconds while anything is queued or running.
- **Filters** — by time (last 24 hours, 7 days, month or all time), who started the job, type and status. Each view has its own address, so it can be bookmarked or shared.
- **Printable error log** — every failure and warning, to print or save as a PDF.

A job failed	The *Conversion jobs* page and its printable error log show the error. Failed jobs can be deleted from *Staged Canvas imports* and re-queued.
Jobs sit in the queue	Expected: only **Maximum simultaneous conversions** of each type run at once, and the rest start as those finish. Raise the limit if the server has spare capacity, and check that cron is running.
Quiz has no questions	Canvas sometimes exports a quiz shell without its item bank. It builds as a hidden placeholder; re-export from Canvas with item banks included.
A URL will not fetch	Check *Security* > *HTTP security* — the host or port may be blocked. Upload the file directly to bypass the server-side fetch.
"Site is being upgraded"	A plugin install or upgrade is running through *Notifications*. Use the command-line upgrade in section 2 to avoid locking the site.
Feature missing	Confirm both plugins are installed and **Allow bulk Canvas import** is ticked. Use Automate 1.4.2+ with CourseUplift 1.3.0+.

CourseUplift and Automate are independent interoperability tools. "Canvas" is a trademark of Instructure, Inc.; these tools are not affiliated with or endorsed by Instructure. Distributed under the GNU GPL v3 or later.